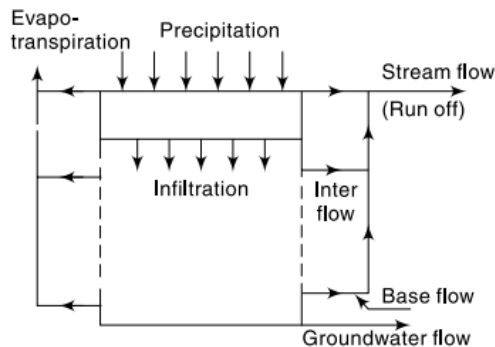


HYDROLOGY

HYDROLOGY

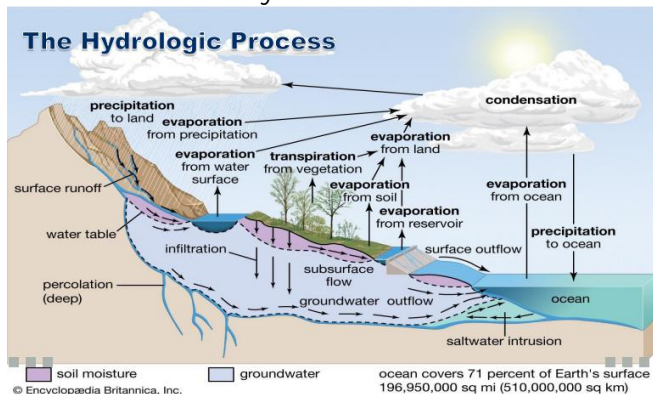
It is the science of water that is concerned with the origin, circulation, distribution and properties of water of the earth



A HYDROLOGIC PROCESS

HYDROLOGIC CYCLE

- “Water cycle”, “water transfer cycle”
- A vertical and horizontal movement of water as either vapor, liquid, or solid between earth’s surface, subsurface, atmosphere, and oceans that occurs continuously in nature.

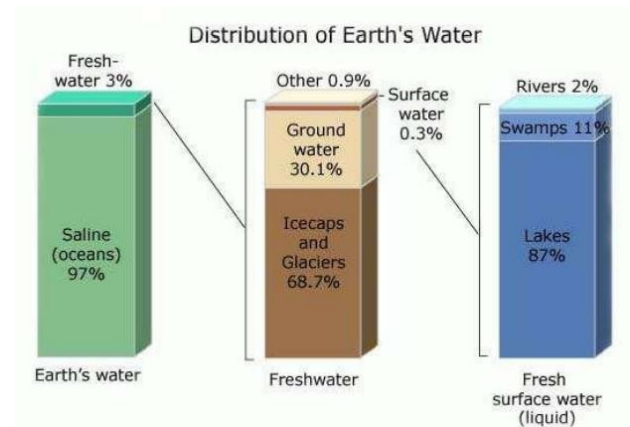


Components of the hydrologic cycle

- Evaporation:** process whereby liquid water is converted to water vapour (vaporization)
- Transpiration:** process of liquid water vaporization/removal contained in plant tissues to the atmosphere through the stomata.
- Condensation:** water vapour becomes liquid.
- Precipitation:** condensation forms clouds which resulted to precipitation. It is the

fall of moisture from atmosphere to the earth’s surface in any form.

- Runoff:** water flowing over the land making its way towards rivers, lakes, oceans, etc as surface or subsurface flow.
- Infiltration:** filtration of water to the inner layers of soil based on its structure.
- Percolation:** vertical movement of water through the soil profile.
- Seepage:** flow of water under gravitational force through permeable layers; lateral movement of water in the soil.



WEATHER

- Short-term state of the atmosphere
- Can change within minutes or hours

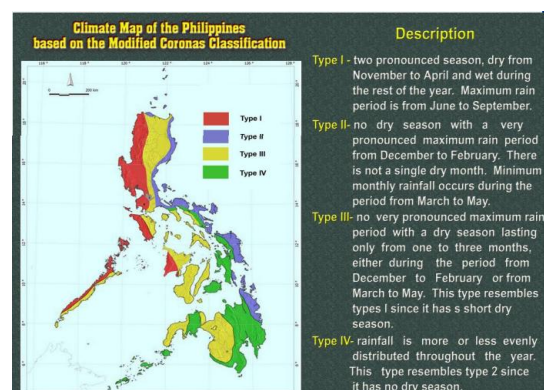
CLIMATE

- Long-term pattern of weather
- Average weather over many years in one specific place.

Climatic Controls for Philippine Climate

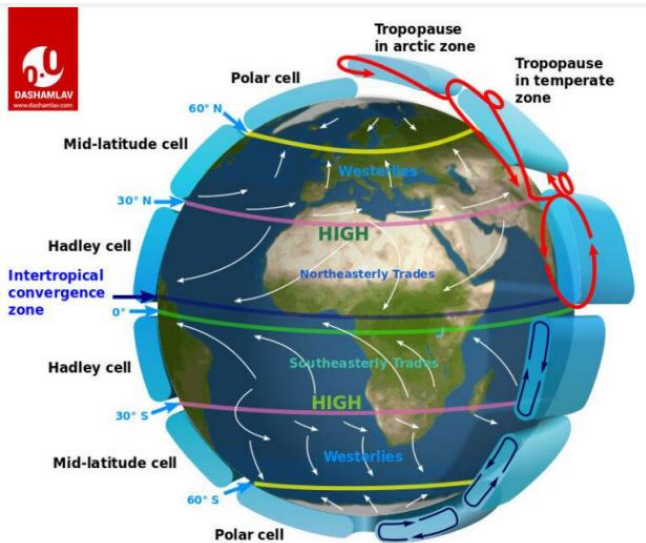
- Topography and location
- Trade winds
- Fronts
- Intertropical Convergence Zone (ITCZ)
- Monsoon winds
- Tropical cyclones
- Easterly waves

Topography and Location



HYDROLOGY

Trade Winds

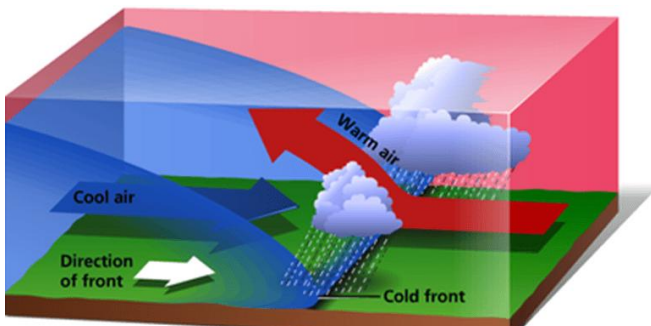


Cold Front

- Boundary along the leading edge of a cold air mass that is pushing out a warm air mass.
- As the cold front passes, the wind changes direction. The weather becomes clear and colder and barometer rises again.
- Type I: observed outside the zone of cyclonic gravity and are generally slow moving.
- Type II: observed within the zone of cyclonic activity and move very rapidly.

Warm Front

- Boundary along a warm air mass that is pushing out a cold air mass. The barometer falls and a long, steady rain begins.
- These are followed by stratus and nimbostratus clouds.
- Temperature rises as warm air replaces cold air, and the barometer stops falling.



Southwest Monsoon (Habagat)

- Winds coming from the southwest.
- Warm and Moist Air
- Moderate to heavy rainfall

- June to October

Northeast Monsoon (Amihan)

- Winds coming from the northeast.
- Cool and Dry Air
- Slight to moderate rainfall
- November to May

MAX WIND SPEED

Tropical depression	30 – 64 kph
Tropical storm	65 – 117 kph
Typhoon	More than 117 kph
Super typhoon	More than 240 kph

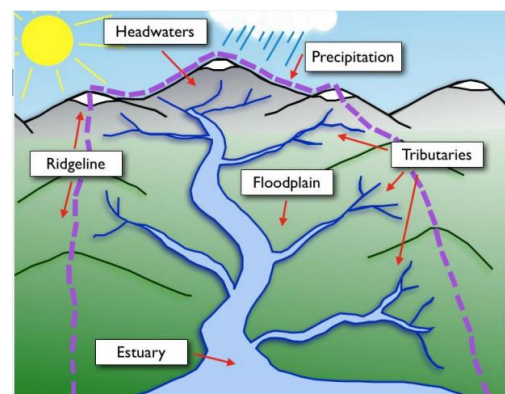
B

WATERSHED

- basic unit of hydrology
- area upstream of a point along a river where all water coming off mainly from precipitation drains towards that point (outlet).

Other terms

- Catchment Area
- Drainage Area
- Catch Basin
- River Basin



Watershed Divide

- “divide”, boundary line along topographic ridges that separates two adjacent watersheds.
- The highest elevation that drains a stream.

Tributaries

- Watershed is made up of tributaries. Stream that flows into a larger stream or other body of water.

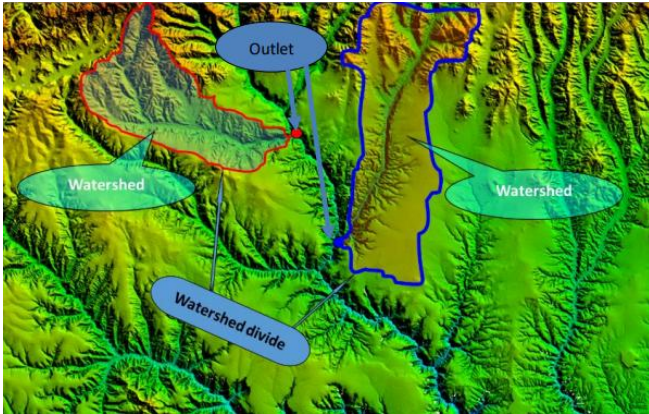
HYDROLOGY

Head Waters

- First order stream. Are the smallest streams at the outer limits of a watershed.

Concentration Point

- "outlet", "measuring point"
- Concentrates as outflow from the basin in the stream channel.



Area Classification

Small watershed: $< 250 \text{ km}^2$

Medium watershed: $250 - 2500 \text{ km}^2$

Large watershed: $> 2500 \text{ km}^2$

WATERSHED CHARACTERISTICS

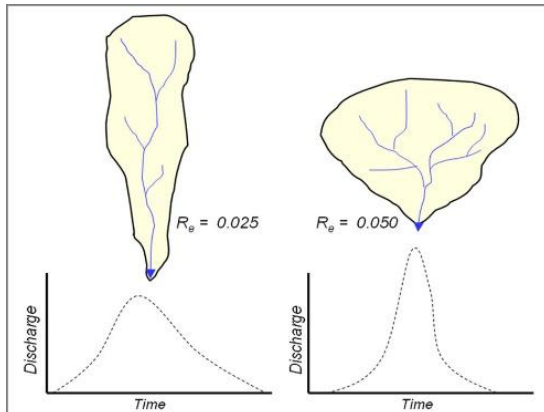
- **Drainage Area:** volume that can be generated from a rainfall.
- **Drainage Basin Length:** distance along the main channel from the watershed outlet to the basin divide.

$$L = 1.312A^{0.568}$$

- **Basin Slope:** rate of change of elevation to the distance along the principal flow path.

$$S = \frac{EL_1 - EL_2}{L}$$

- **Basin Shape:** reflects how runoff will be collected at the outlet.



C

PRECIPITATION

- that part of atmospheric moisture that falls on the earth surface. It is primary input into the hydrologic cycle.
- condensation or sublimation of atmospheric moisture.

This transformation may be brought about by one or more of the following causes:

1. adiabatic cooling
2. mixing of air masses of varying temperature
3. radiation cooling, and
4. contact cooling.

Type	Intensity
1. Light rain	trace to 2.5 mm/h
2. Moderate rain	2.5 mm/h to 7.5 mm/h
3. Heavy rain	> 7.5 mm/h

FORMS OF PRECIPITATION

Drizzle: a light steady rain in fine drops, about 0.5 mm and intensity less than 1 mm/hr.

Rain: is the condensed water vapor of the atmosphere falling in drops ($> 0.5 \text{ mm}$, maximum size is 6 mm) from the clouds.

Glaze: is the freezing of drizzle or rain when they come in contact with cold objects.

Sleet: is the frozen rain drops which falling through air at subfreezing temperature.

Snow: is the ice crystal resulting from sublimation; i.e., water vapor condenses to ice.

is iced crystals fused together.

Hail: refers to small lumps of ice (3.5 mm diameter formed by alternate freezing and melting, when they are carried up and down in highly turbulent air currents.

Dew: is moisture condensed from the atmosphere in small drops upon cool surfaces

Frost: is a feathery deposit of ice formed on the ground or on the surface of exposed object by dew or water vapor that has frozen

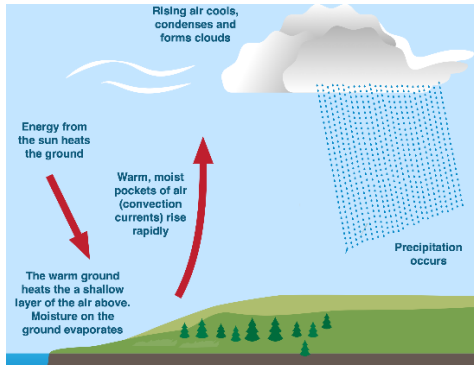
Fog: is a thin cloud of varying sizes formed at the surface of the earth by condensation of atmospheric vapor, usually interfering with visibility.

Mist: is a very thin fog.

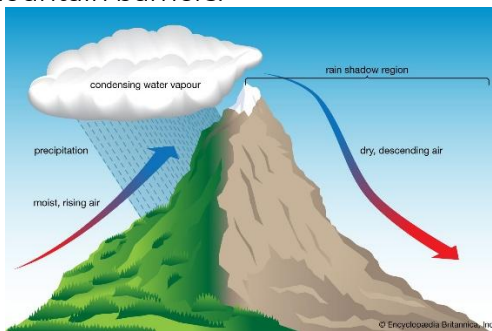
TYPES OF PRECIPITATION

1. **Convective Precipitation:** results from the differential heating of air masses near the ground surface resulting in the upward movement of warmer air masses.

HYDROLOGY

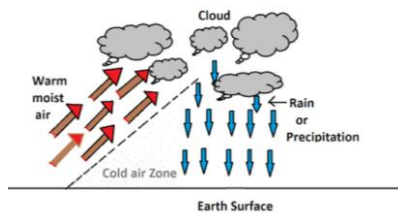


2. **Orographic Precipitation:** results from mechanical lifting of the air mass over mountain barriers.

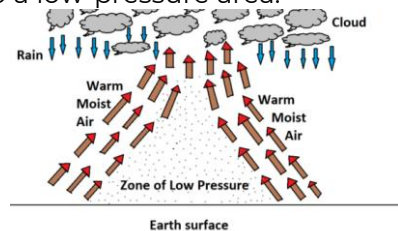


3. **Cyclonic Precipitation:** are associated with the movement of air masses due to differences in barometric pressure. is triggered by the presence of a low pressure into which air flow converges resulting in the lifting and subsequent cooling of air masses.

- a. **Frontal:** lifting of warm air over cold air at the contact zone between air masses having different characteristics



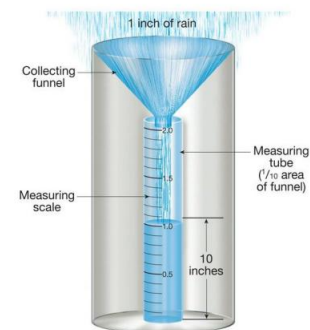
- b. **Non-frontal:** air is lifted through horizontal convergence of the inflow into a low-pressure area.



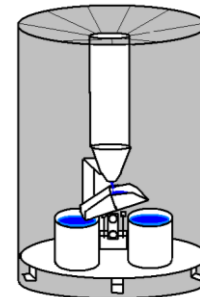
MEASUREMENT OF PRECIPITATION

Standard Rain Gauge: a non-recording weather instrument used to measure rainfall. "8-inch rain gauge"

The standard rain gauge



Tipping Bucket Rain Gauge: measures rainfall through the collection and counting of droplets.



Weighing-Bucket Type: the catch from the funnel empties into a bucket mounted on a weighing scale.

Natural-Syphon Type: "float-type gauge"; rainfall collected by a funnel-shaped collector is led into a float chamber causing a float to rise.

Telemetering Raingauges: recording type and contain electronic units to transmit the data on rainfall to a base station both at regular intervals and on interrogation.

Radar Measurement of Rainfall: measuring areal extent, location, and movement of rain storm.

ESTIMATION OF AVERAGE AREAL PRECIPITATION

1. **Arithmetic Mean:** averaging arithmetically all the rainfall depths.

$$P_{av} = \frac{P_1 + P_2 + P_3 + \dots + P_n}{n} = \frac{\sum_{i=1}^n P_i}{n}$$

2. **Thiessen Polygon Method:** assumes that the rainfall in an area may be taken as similar to that recorded in the nearest gaging station.

HYDROLOGY

$$P = \frac{A_1 P_1 + A_2 P_2 + A_3 P_3 + \dots + A_n P_n}{A_1 + A_2 + A_3 + \dots + A_n}$$

- 3. Isohyetal Method:** the rainfall depths recorded in all the stations in and around an area of interest are plotted on a map of desirable scale.

$$\bar{P} = \frac{a_1 \left(\frac{P_1 + P_2}{2} \right) + a_2 \left(\frac{P_2 + P_3}{2} \right) + \dots + a_{n-1} \left(\frac{P_{n-1} + P_n}{2} \right)}{A}$$

- 4. Grid-Point Method:** computer-aided.
- 5. Inverse Distance Ratio Method:** rolling areas and non-uniform rain gauges.

ADEQUACY OF RAINFALL STATIONS

$$N = \left(\frac{C_v}{\varepsilon} \right)^2 \quad \text{optimal number of stations}$$

$$C_v = \frac{100 \times \sigma_{m-1}}{\bar{P}} \quad \text{coefficient of variation}$$

$$\sigma_{m-1} = \sqrt{\left[\frac{\sum_{i=1}^m (P_i - \bar{P})^2}{m-1} \right]} \quad \text{standard deviation}$$

P_i = precipitation magnitude in the i^{th} station

$$\bar{P} = \frac{1}{m} \left(\sum_{i=1}^m P_i \right) \quad \text{mean precipitation}$$

RAINFALL ADJUSTMENTS

- 1. Estimation of Missing Rainfall Data**

- a. Arithmetic Mean Method**

$$P_x = \frac{1}{m} (P_1 + P_2 + \dots + P_m)$$

within 10% of the normal rainfall

- b. Normal Ratio Method**

$$P_x = \frac{N_x}{m} \left(+ \frac{P_1}{N_1} + \frac{P_2}{N_2} + \dots + \frac{P_m}{N_m} \right)$$

more than 10% of the normal rainfall

- 2. Double Mass Analysis:** used to check the consistency of a rain gauge record

$$k = \frac{\text{gradient of line before change}}{\text{gradient of line after change}}$$

HYDROLOGIC FREQUENCY ANALYSIS

- 1. Deterministic Process:** one in which a definite law of certainty exists
- 2. Probabilistic Process:** governed by chance; time series INDEPENDENT
- 3. Stochastic Process:** governed by chance; time DEPENDENT

Recurrence Interval or Return Period (T)

- Occur once,** may be equaled or exceeded.

$$P = \frac{1}{T} \quad P = 1 - \frac{1}{T}$$

- NOT occur** in any series of N years

$$P_{\text{non-exceedance}} = \left(1 - \frac{1}{T} \right)^N$$

- Occur** in any series of N years

$$P_{\text{exceedance}} = 1 - \left(1 - \frac{1}{T} \right)^N$$

- Occurring **2 times** in N successive years

$$P_{2,n} = \frac{n!}{(n-2)! 2!} P^2 q^{n-2}$$

Frequency Distributions

- 1. Normal Distribution:** data is more or less normally distributed
- 2. Log-Normal Distribution:** used when there is skewness in the data distribution
- 3. Pearson Type III Distribution:** considers further the skewness of the logarithmically transformed hydrologic data; preferred for flood flow frequency; log-normal distribution is a special case of this method
- 4. Gamma Density Function:** gives the best fit for most stations for 1-week, 2-week, 3-week and monthly rainfall totals of the country

D

EVAPOTRANSPIRATION

- Predicting the irrigation requirements of crops
- Estimating runoff from river basins
- Determining the amount of water available from storage reservoir

ESTIMATING EVAPOTRANSPIRATION

Indirect Method

Pan Evaporation Method

$$ET_o = k_p E_{pan}$$



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Lysimeters: measure the amount of actual evapotranspiration which is released by plants, usually crops or trees



Empirical Equation

Penman Equation: combined energy balance and mass transfer to compute evaporation from open water surfaces.

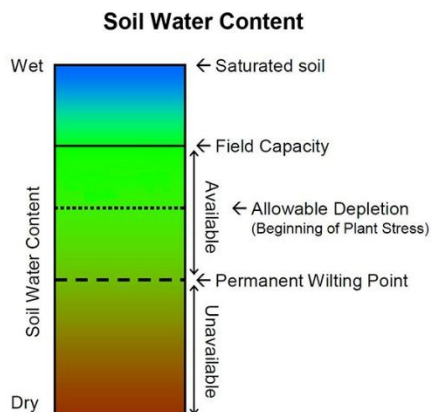
$$\lambda ET = \frac{\Delta(R_n - G) + \rho_a c_p \frac{(e_s - e_a)}{r_a}}{\Delta + \gamma \left(1 + \frac{r_s}{r_a}\right)}$$

Penman - Monteith Equation: combination method extended it to cropped surfaces. Includes factors to account for stomatal resistance in plants and evaporation from the soil surface.

$$ET_o = \frac{0.408\Delta(R_n - G) + \gamma \left(\frac{C_n}{(T + 273.16)} \right) u_2 (e_s - e_a)}{\Delta + \gamma (1 + C_d u_2)}$$

Hydrologic Budget

$$P - R_s - G_o - E_{act} = \Delta S$$



EVAPORATION

- Liquid changes to gaseous state at the free surface, below the boiling point through the transfer of heat energy.
- The rate of evaporation is dependent on:
 - Vapour pressure

$$E_L = C(e_w - e_a)$$
 - Air and water temperature
 - Wind speed
 - Atmospheric pressure
 - Water quality
 - Size of the water body

Evaporimeters

Class A Evaporation Pan: standard pan 1210 mm diameter and 255 mm depth.

ISI Standard Pan: modified Class A Pan 1220 mm diameter and 255 mm depth.

Colorado Sunken Pan: 920 mm square and 460 mm deep

US Geological Survey Floating Pan: 900 mm side and 450 depths.

Water-Budget Method

$$P + V_{is} + V_{ig} = V_{os} + V_{og} + E_L + \Delta S + T_L$$

Energy-Budget Method

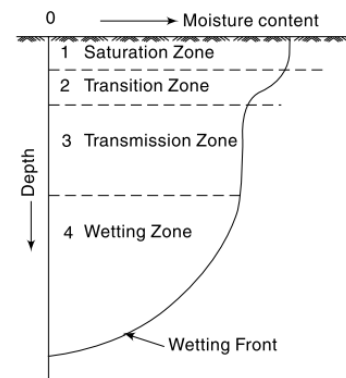
$$H_n = H_a + H_e + H_g + H_s + H_i$$

TRANSPIRATION

- Water leaves the body of a living plant and reaches the atmosphere as water vapour.

E INFILTRATION

- the passage of water into the soil surface. In general, infiltration has a high initial rate that diminishes with time.



STAGE SEQUENCE

- Surface Entry
- Filling up of soil profile storage potential
- Movement or transmission of water within the soil profile

FACTORS AFFECTING INFILTRATION

- Fluid. viscosity, turbidity, and depth of water
- Medium. Porosity, structure, structure stability, moisture holding capacity, antecedent soil moisture, mineralogy
- Vegetation
- Cultural Practices
- Topography

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MEASUREMENT OF INFILTRATION

1. Infiltrometers

- Flooding Type: double-ring infiltrometers
- Rainfall Simulator Type: installed over a small plot.

2. Basin Method

Large area is ponded. Not practical and difficult to measure the infiltration rates.

3. Watershed Hydrograph Method

Involves hydrograph analyses to separate surface runoff from total streamflow

INFILTRATION EQUATION

1. Lewis-Kostiakov

The equation will only apply for soil underlain with impervious layers or for small t values in soils with pervious layers.

$$i = ct^a \quad ; 0 < a < 1$$

$$f = act^{a-1}$$

2. Horton Equation

Decay type function where continuous infiltration and wetting of soil decreases due to decrease in soil profile potential.

$$i = f_c + \frac{(f_0 - f_c)}{k}(1 - e^{-kt})$$

$$f = f_c + (f_0 - f_c)e^{-kt}$$

3. Philip

Vertical infiltration in terms of the potential differential equation. Very accurate for largest values.

$$i = St^{1/2} + At$$

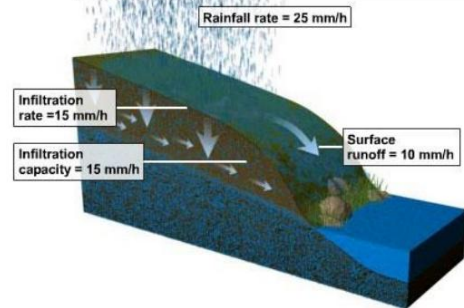
$$f = \frac{S}{2}t^{-1/2} + A$$

4. Green-Ampt Equation

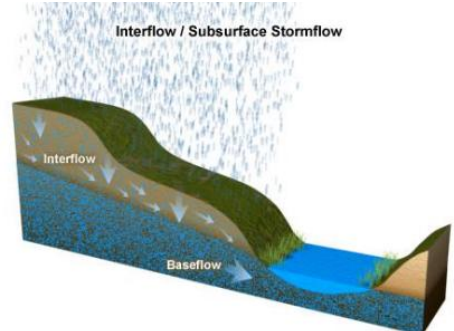
Infiltration capacity based on Darcy's law

$$f_p = K \left(1 + \frac{\eta S_c}{F_p} \right) \quad \text{or could be} \quad f_p = m + \frac{n}{F_p}$$

Relationship between Rainfall, Infiltration, and Runoff (Rainfall Rate > Infiltration Capacity)

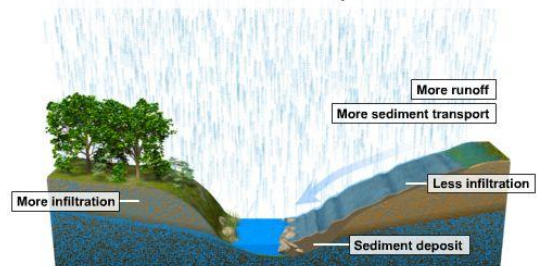


- Interflow:** part of precipitation which infiltrates into the soil surface and moves laterally through the upper soil horizons towards streams as perched groundwater above the major groundwater level.



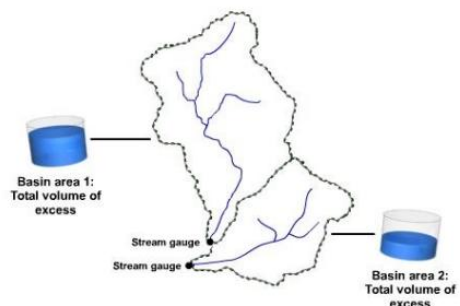
- Groundwater Runoff:** groundwater discharge into stream due to deep percolation of the infiltrated water into the groundwater aquifers.

Influences of Deforestation on Runoff, Groundwater, and Sediment Transport



Influence of Basin Area on Runoff Volume

when rain falls in a uniform manner over a larger basin and a smaller basin, the larger basin produces more runoff volume.



F

RUNOFF

- processes and pathways through which excess water becomes streamflow. It is that part of precipitation which appears in surface streams or rivers.

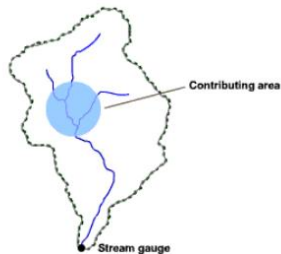
RUNOFF COMPONENTS

- Surface Runoff:** which travels over the ground surface and through the channels to reach the basin outlet.

HYDROLOGY

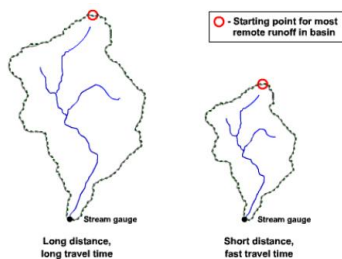
Contributing Area of Storm Determines Runoff

Many storms will cover only part of a basin. For most situations, the runoff volume will be determined by the contributing area — that part of the basin covered by a storm — not the total size of a basin.



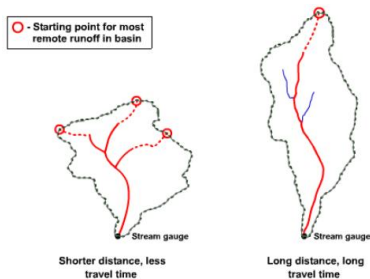
Influences of Basin Sizes on Runoff

Runoff traveling from most upstream point of the larger basin will travel a longer path, and therefore take longer to reach the basin outlet than runoff traveling from the farthest point in the smaller basin.



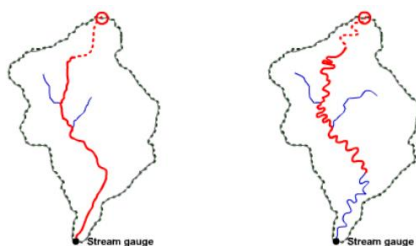
Influence of Basin Shape on Runoff

The runoff in the rounder basin will arrive more quickly at the basin outlet



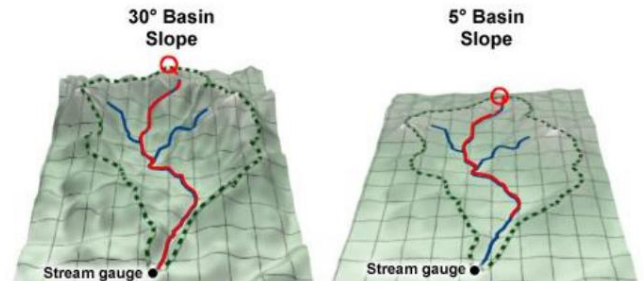
Influence of Straight vs Meandering Flowpath on Runoff

Meanders increase the travel time of runoff through the basin, and may reduce the overall runoff volume



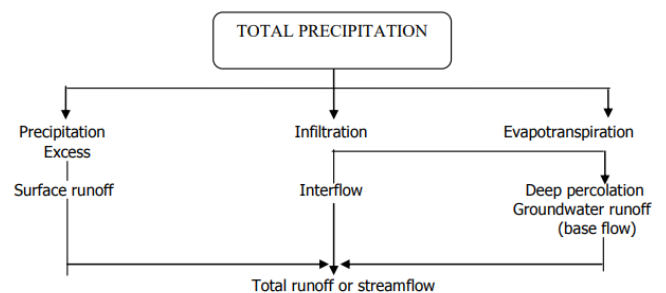
Influence of Basin Slope on Runoff

As the ground becomes increasingly steep, water will move faster and will have less time in contact with the ground surface, reducing the time during which it could infiltrate



Other useful definitions referring to runoff components include:

- Overland flow** – that part of surface runoff that flows over the surface towards stream channels; same as surface runoff
- Direct runoff** – same as surface runoff
- Subsurface runoff** – same as interflow
- Baseflow** – same as groundwater runoff.
- Precipitation excess or runoff coefficient** – that part of precipitation that contributes directly to surface runoff.
- Channel interception** – rain that falls directly into the stream channel.
- Time to peak** – time between the center of mass for a rainfall event and the peak streamflow rate.
- The **stage**, the **gage height**, and the **level of the water surface** all are considered to have the same meaning.



FACTOR AFFECTING RUNOFF

a. Climate

1. Precipitation

- Form
- Intensity
- Duration
- Time distribution
- Areal distribution
- Direction of storm movement in relation to the orientation of the watershed
- Frequency of occurrence

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- Antecedent precipitation- an index of antecedent soil moisture condition.

2. Vegetation interception of rainfall

3. Evapotranspiration

b. Physiographic Factors

1. Basin characteristics

o Geometric factors

- Size
- Shape
- Slope
- Orientation
- Elevation
- Stream density

o Physical factors

- Land use and cover
- Surface infiltration conditions
- Geologic conditions (permeability, aquifer transmissivity and storage, coefficient)
- Topographic conditions (depression storage, lakes, swamps)
- Artificial drainage
- Man-made modifications, dams, terraces, etc

c. Channel characteristics

o Discharge capacity

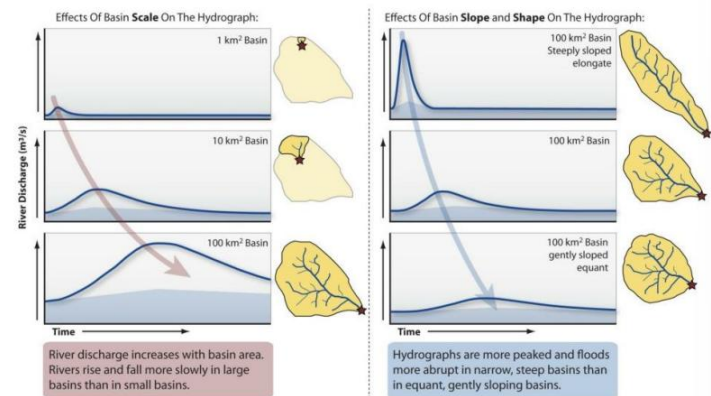
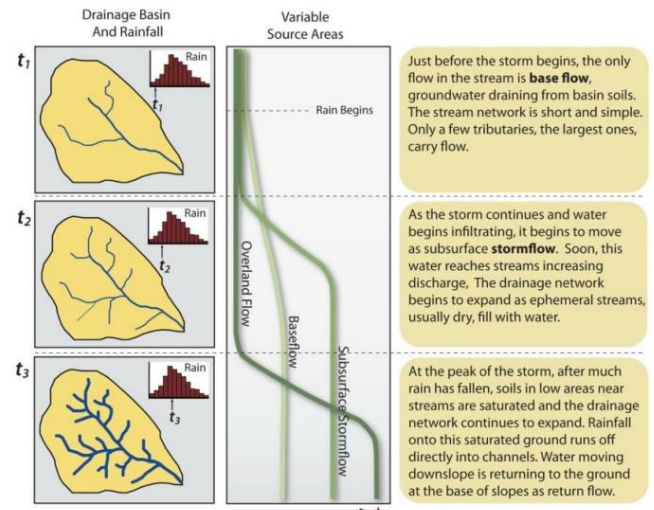
- Cross section
- Slope
- Roughness
- Length
- Tributaries

o Storage capacity (backwater effects)

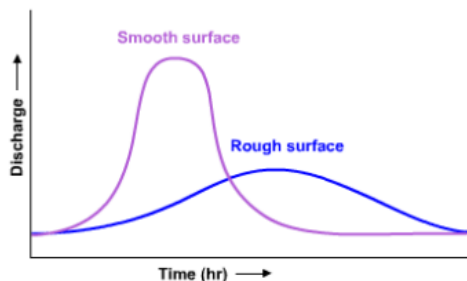
Channel storage: has a prominent effect on runoff in **very large watersheds**.

Land use influence: the predominant factor affecting runoff in **smaller watersheds**.

Small watershed: (<250 sq. km area) very sensitive to high intensity rainfall of short duration.

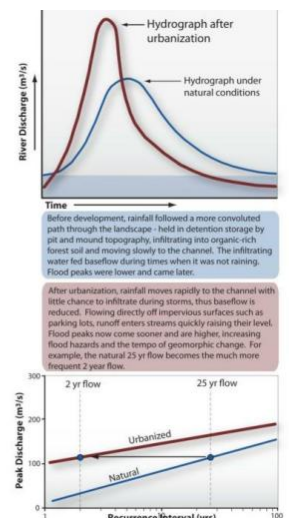
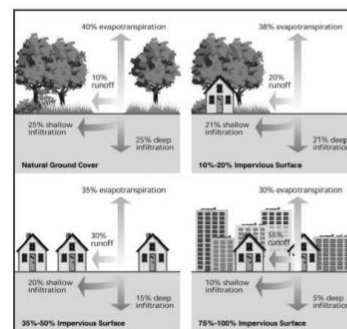


Hydrographs for Rough vs Smooth Channels



- The greater the roughness, the more turbulent the flow.
- More turbulent flow results in slower runoff and streamflow velocities.
- Reducing channel roughness results in faster streamflow velocities and greater peak flows.

Watershed Urbanization

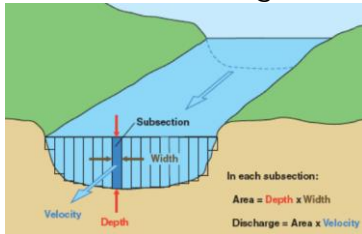


HYDROLOGY

RUNOFF MEASUREMENT AND ESTIMATION

a. Direct Method

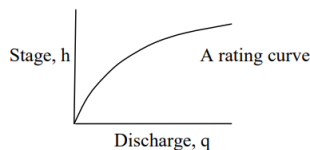
- **Current-meter:** determine the discharge in each subsection of a channel cross section and summing the subsection discharges.



$$V = a + bN$$

b. Indirect Method

- Orifice, weir, flume, and other control structure
 - Weirs: $q = CH^a$
- **Staff gauges and water level recorders:** These are used to measure the stage or depth of flow above a datum or reference level.



c. Use of Empirical Formula

- **Rational Equation**

$$q_p = CiA$$
- **Kirpich Formula [Time of concentration (tc)]**
required time for a water particle to flow hydraulically from the most distant point in the watershed to the outlet

$$t_c = 0.0078L^{0.77}S^{-0.385} \quad ; L - ft$$

$$t_c = 0.02L^{0.77}S^{-0.385} \quad ; L - m$$
- **SCS Lag Formula [Time of concentration (tc)]**

$$t_c = 0.0134L^{0.8} \left(\frac{1000}{CN} - 10 \right)^{0.7} S^{-0.05}$$
- **Soil Conservation Service (SCS) Method/ Curve Number**

$$Q = \frac{(I - 0.2S)^2}{(I + 0.8S)} \quad S = \frac{25,400}{CN} - 254$$

Low CN (30-60)- low runoff (sandy soil, forest)
High CN (70-100)- high runoff (clay, paved area)

Gumbell Distribution

$$X_T = M + S(0.78 \ln T - 0.45)$$

G

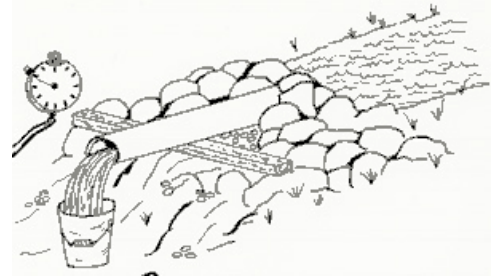
STREAMFLOW

BASIC EQUATION

$$Q = AV$$

METHODS OF FLOW MEASUREMENT

1. Volumetric and Gravimetric Method



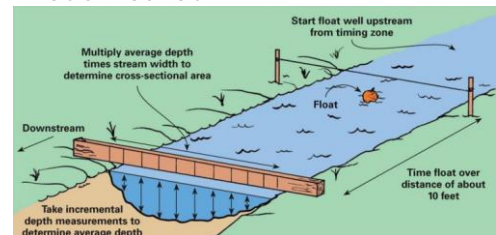
2. Slope-Area Method

$$V = \frac{k}{n} R^{2/3} S^{1/2}$$

k=1-metric, 1.486-english

3. Velocity-Area Method

• Float Method



• Current Meter Method

$$V = a + bN$$

Types

Anemometer and Propeller
 Electromagnetic
 Doppler Type
 Optical Strobe Velocity Meter

4. Control-Section Method

Various Types of Commonly Used Flow Measurement Structures

Table 1. Types of Weir, Description and Discharge Evaluation

Classification	Description	Discharge Evaluation
SHARP-CRESTED/ THIN-PLATE - weir constructed with a crest of vertical thin plate		
Contracted Rectangular		$Q = 1.84 (L - 0.2H) H^{3/2}$ Q = discharge, m³/s L = length of crest, m H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m
Suppressed Rectangular		$Q = \frac{2}{3} C_d L H \sqrt{2gH}$ $C_d = 0.615 \left(1 + \frac{1}{H + 1.6} \right) \left[1 + 0.5 \left(\frac{H}{H + D} \right)^2 \right]$ Q = discharge, m³/s C_d = discharge coefficient L = length of crest, m g = gravitational acceleration, m²/s H = head, m (mm for computing μ) D = distance from the crest to the bottom of the approach channel, mm

HYDROLOGY

Cipoletti Contracted		$Q = 1.86 LH^{3/2}$ Q = discharge, m^3/s L = length of crest, m H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m
V-notch		$Q = \frac{8}{15} \sqrt{2g} C_d H^{5/2}$ Q = discharge, m^3/s g = gravitational acceleration, m^2/s^2 C_d = discharge coefficient H = head (difference between the elevation of the weir crest and the elevation of the water surface in the weir pool), m

Table 2. Types of Flumes, Description and Discharge Evaluation

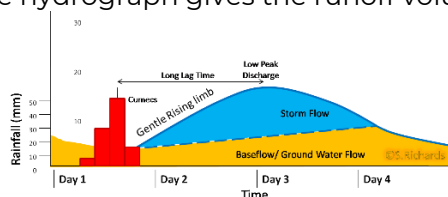
Long-throated (Rectangular control section)		$Q = C_d C_v \frac{2}{3} \left(\frac{2}{3} g \right)^{0.50} b_1 H_1^{1.50}$ Q = discharge, m^3/s g = gravitational acceleration, m^2/s^2 $H_1 = (3/2)y_c$ y_c = total upstream energy head over the weir crest, m C_d = coefficient of discharge C_v = coefficient due to approach velocity
Cut-throat		$Q = CH_u^n$ $C = KW^{1.025}$ Q = discharge, m^3/s C = free flow coefficient H_u = upstream gauge reading (2L/9 from throat) K = flume length coefficient W = throat width, m
Parshall		$Q = kH_u^n$ Q = discharge, m^3/s k = dimensional factor which is a function of throat width H_u = upstream gauge reading $n = 1.522$ to 1.60

Table 3. Types of Orifices, Description and Discharge Equation

Circular Sharp-Edged		$Q = C_d C_v A \sqrt{2g(h_1 - h_2)}$ $Q = C_d C_v A \sqrt{2g\Delta h}$ Q = discharge, m^3/s C_d = discharge coefficient C_v = velocity coefficient A = area of the orifice, m^2 $h_1 - h_2$ = head differential across the orifice Δh = upstream head above the center of the orifice
Rectangular Sharp-Edged		

H HYDROGRAPHS ANALYSIS

- A hydrograph is a graphical representation of the instantaneous runoff rate against time. The area under the hydrograph gives the runoff volume.

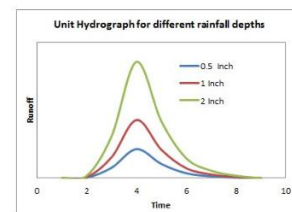


Runoff Hydrograph Properties

- Peak Flow (qp):** determine magnitude of floods
- Time to peak:** watershed response time
- Runoff volume (V):** total water yield from a storm for a given watershed
- Recession Time (tr):** time for surface and interflow to recede; duration of flooding
- Base time (tb):** total time to peak and recession time
 $tb = tp + tr$
- Base flow:** low flow, dependable flow, groundwater discharge

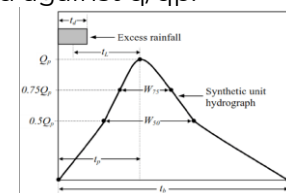
Unit Hydrograph

A direct runoff hydrograph resulting from one unit of constant intensity uniform rainfall occurring over the entire watershed.



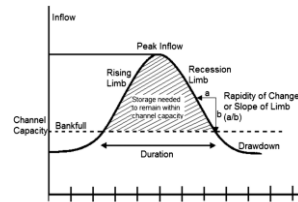
Synthetic Hydrograph

a dimensionless hydrograph made from natural or unit hydrograph in which the time to peak (tp) and the peak runoff rate (qp) are considered; and t/tp is plotted against q/qp.



Basic Hydrograph

It assumes that hydrographs from small watersheds have similar forms. It is established by plotting 100 arbitrary units of flow and 100 arbitrary units of time



Triangular Hydrograph

It is an approximation of the basic hydrograph. It was developed mainly to simplify flood routing procedures.

